

Olist E-Commerce Marketplace Analysis

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Exploratory Data Analysis and Business Insights Report

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Executive Summary

This project analyzes the Olist e-commerce dataset to extract operational and commercial insights that support data-driven decisions. I conducted a full exploratory analysis in Python (cleaning, merging, feature engineering, geographic patterns, and review text mining) and built a multi-page Power BI dashboard (Overview, Customers, Sellers, Deliveries, Reviews). This report summarizes the main findings and recommendations, accompanied by a concise slide deck.

Key Findings

- **Orders & Revenue:** solid 2017 growth with a strong Black Friday spike; revenue concentrated in a few categories and sellers.
- **Customer Behavior:** retention is low — nearly all customers buy once; repeat buyers are 3%.
- **Delivery & Satisfaction:** delivery performance is the main driver of ratings (4.2 stars on-time vs. 2.5 late).
- **Geography:** distance from logistics hubs increases late and “never arrived” deliveries.
- **Seller Concentration:** top 10% of sellers represent 68% of orders; high volume does not guarantee better CX.
- **Reviews:** negative keywords focus on logistics (“didn’t receive”, “late delivery”); positive reviews highlight quality and fast delivery.

Recommendations

- **Strengthen logistics:** expand or optimize regional hubs, negotiate performance-based SLAs, and reduce late/failed deliveries.
- **Proactive communication:** implement early alerts for delayed shipments and improved tracking transparency to mitigate low ratings.
- **Seller performance management:** track seller-level KPIs (on-time rate, never-arrived rate, claim rates) and provide operational playbooks for underperformers.
- **Event planning:** prepare temporary capacity for peak periods (Black Friday etc.) to maintain delivery performance under high demand.
- **Voice-of-customer automation:** implement keyword-based alerts for critical complaint patterns (“didn’t receive”, “wrong product”, “late”) to prioritize rapid intervention.
- **Customer retention:** launch targeted nudges or incentives to encourage second purchases, addressing the extremely low return rate.

Deliverables

- A fully reproducible Python notebook containing data cleaning, EDA, geospatial insights, and review text analysis.
- A multi-page Power BI dashboard (Overview, Customers, Sellers, Deliveries, Reviews).
- This detailed written report, summarizing insights and recommendations.
- A concise executive slide deck for quick presentation.

1 Project and Data Overview

This project is based on the Brazilian E-Commerce Public Dataset released by **Olist**, one of the largest marketplace integrators in Brazil. Olist connects small and medium sellers to major online marketplaces, providing a unified platform for catalog management, logistics coordination, and customer experience. The dataset captures real commercial activity from 2016 to 2018 and covers the full customer journey — orders, payments, product attributes, delivery performance, geolocation, and written reviews. Personal and corporate names were anonymized by the providers (including substitutions using Game of Thrones house names).

The data is publicly available on Kaggle:

Brazilian E-Commerce Public Dataset by Olist <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Included Data Sources

The full dataset is composed of **nine relational tables**, each representing a different layer of the e-commerce pipeline:

- **orders** — order-level details including timestamps and status.
- **order_items** — products, quantities, seller IDs, and freight costs.
- **order_payments** — payment methods, number of installments, and amounts.
- **order_reviews** — customer-written reviews and ratings.
- **customers** — customer IDs and location attributes.
- **sellers** — seller IDs and geographic information.
- **products** — product category, name lengths, and dimensional attributes.
- **geolocation** — latitude/longitude coordinates associated with Brazilian zip codes.
- **product_category_name_translation** — mapping between Portuguese and English product category names.

Altogether, these datasets offer a multi-dimensional view of marketplace operations, enabling analysis of demand trends, logistics efficiency, customer satisfaction, and seller performance.

Data Schema

The relationships between tables follow a typical star-schema structure centered on the order ID, connecting customers, sellers, products, and reviews. Figure 1 illustrates the full schema used throughout the analysis.

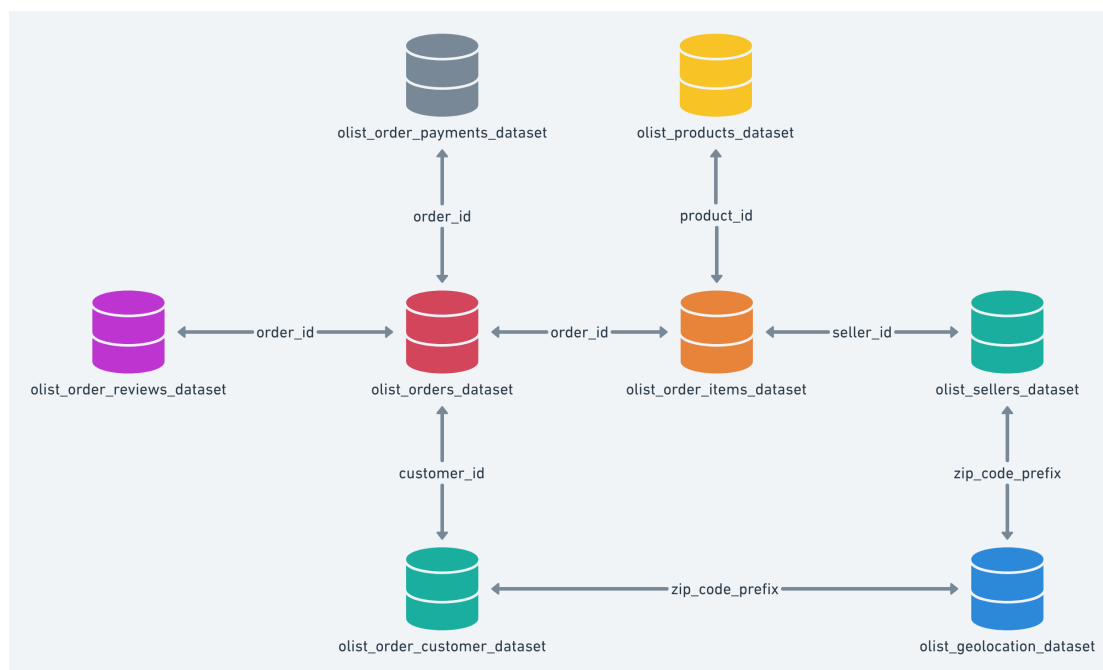


Figure 1: Olist Dataset Relational Schema

2 Data Overview

After loading and merging the nine Olist datasets, I performed an initial diagnostic review to understand the scale, completeness, and distribution of the data. The combined dataset covers a substantial volume of commercial activity across multiple marketplaces.

Dataset Size and Key Entities

The cleaned and consolidated data contains:

- **99,441 orders**
- **96,096 unique customers**
- **3,002 unique sellers**
- **99,224 customer reviews**

These entities span a nationwide customer and seller base, enabling meaningful analysis of demand patterns, delivery performance, geographic disparities, and marketplace behavior.

Temporal Coverage and Selection

The raw dataset covers the period from September 2016 to October 2018. However, the earliest and latest months contain incomplete or extremely sparse data:

- **Sep–Dec 2016:** very low volume due to ramp-up in data collection.
- **Sep–Oct 2018:** sharp drop in order counts (only 16 and 4 orders), indicating incomplete ingestion.

To ensure statistical reliability and avoid misleading trends, I defined a focused and consistent analysis window:

Analysis Period Used: January 2017 to August 2018. This 20-month period represents the most complete and stable portion of the dataset and is used throughout the EDA, KPIs, and dashboard.

Data Quality Notes

Within the selected date range, the data is sufficiently dense and stable to support:

- Month-over-month demand and revenue analysis.
- Customer and seller segmentation.
- Delivery-time and logistics performance evaluation.
- Review-based sentiment and complaint signal extraction.

By constraining the analysis to a consistent data window, the results reflect real operational patterns rather than artifacts from incomplete collection.

3 Analytical Approach

This section outlines the analytical structure used to transform the Olist dataset into business insights. The focus is on understanding customer behavior, sales performance, delivery efficiency, and customer satisfaction within the Brazilian e-commerce landscape.

3.1 Analytical Objectives

The analysis was guided by four core objectives:

- **Understand sales and order dynamics** across time, identifying seasonality, growth phases, and notable events such as Black Friday.
- **Profile customer behavior** to uncover purchasing patterns, geographic distribution, and retention characteristics.
- **Evaluate delivery performance**, measuring punctuality, delays, and logistics bottlenecks.
- **Assess customer satisfaction** using review scores and text-based indicators to identify strengths and recurring pain points.

These objectives shaped the indicators and dashboards developed throughout the project.

3.2 KPI and Metric Construction

All performance indicators were designed to be transparent, replicable, and aligned with standard e-commerce analytics. The key metrics include:

- **Revenue and Order Value Metrics:** Gross revenue, average order value (AOV), payment installments, and freight share.
- **Customer Metrics:** Total unique customers, repeat purchase rate, geographic distribution, and cohort-based activity patterns.
- **Delivery Metrics:** On-time vs. delayed delivery rate, average delivery time, and distance effects by state and city.
- **Review and Satisfaction Metrics:** Review score distribution, percentage of low (1–2) vs. high (4–5) scores, and sentiment tendencies from review comments.

The KPI logic was fully integrated into the Power BI data model to ensure consistent evaluation across dashboards.

3.3 Analytical Workflow

The analytical work followed three structured phases:

1. **Exploratory Data Analysis (EDA):** Identification of distribution patterns, data inconsistencies, behavioral trends, and relationships across the nine datasets.
2. **Data Modeling and Integration:** Construction of a unified analytical model connecting orders, customers, sellers, products, geolocation, payments, delivery performance, and review data. A Date Dimension was added to support time intelligence and comparisons.
3. **Dashboard Development and Insight Consolidation:** Creation of interactive Power BI dashboards organized into thematic sections (Overview, Customers, Sellers, Delivery Performance, Reviews). Insights were then compiled into this report and into the final presentation.

4 Key Business Insights

4.1 Orders & Revenue Dynamics

The Olist marketplace shows clear growth throughout 2017 and mid-2018, supported by both an expanding order base and rising revenue. The following chart summarizes monthly performance using the validated analysis period (Jan 2017–Aug 2018). It reveals steady growth, moderate seasonal fluctuations, and a pronounced revenue spike during major retail events.

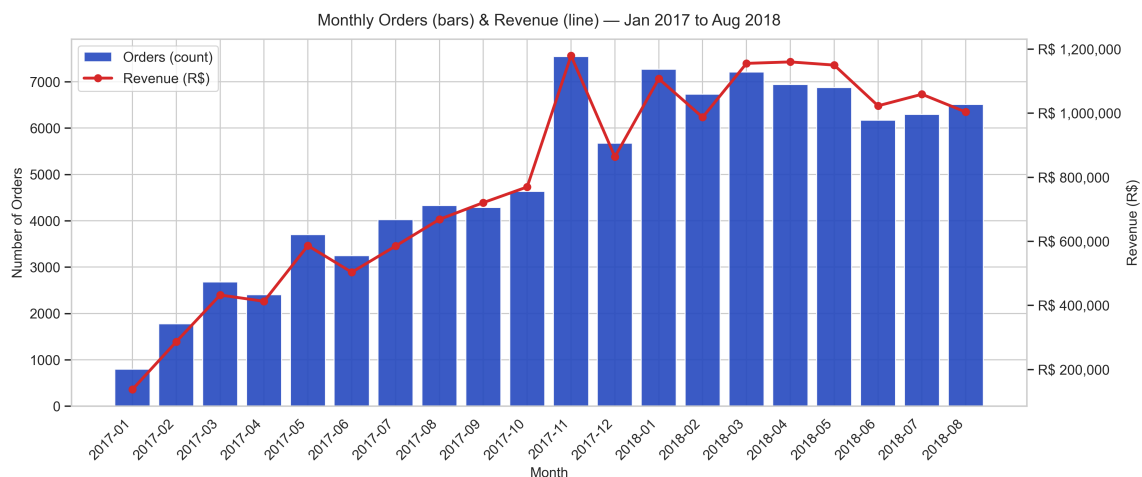


Figure 2: Monthly Orders and Revenue (Jan 2017–Aug 2018)

To better illustrate short-term fluctuations and identify event-driven peaks, daily order volume was also analyzed. The graph below highlights a pronounced surge in late November 2017, corresponding to **Black Friday**, which is the largest shopping event in Brazil and generates a sharp, short-term uplift in sales activity.

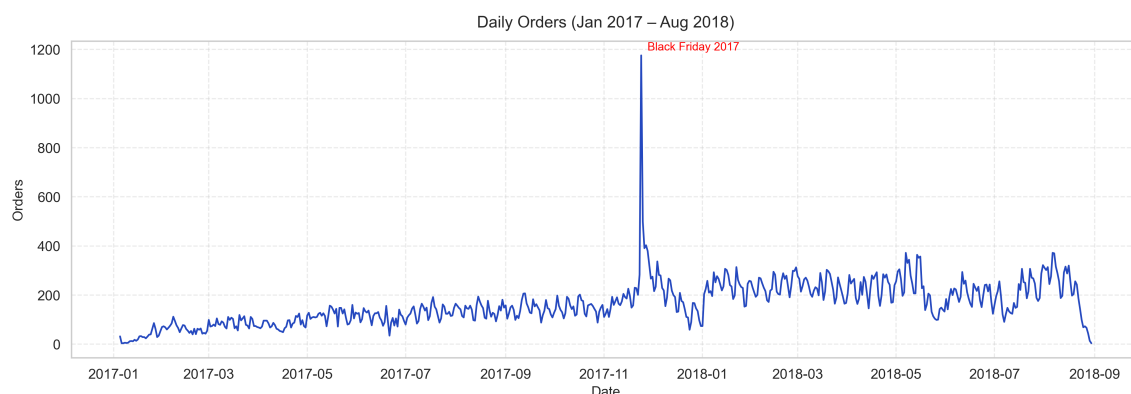


Figure 3: Daily Orders — Highlighting the Black Friday Spike (Nov 2017)

Key insight: Black Friday drives the strongest demand peak in the dataset, compressing a month's worth of orders into just a few days. This has clear implications for operational capacity planning, logistics preparation, and seller coordination.

4.2 Payment Methods

This section summarizes how customers pay on the marketplace and whether payment method correlates with high-value orders.

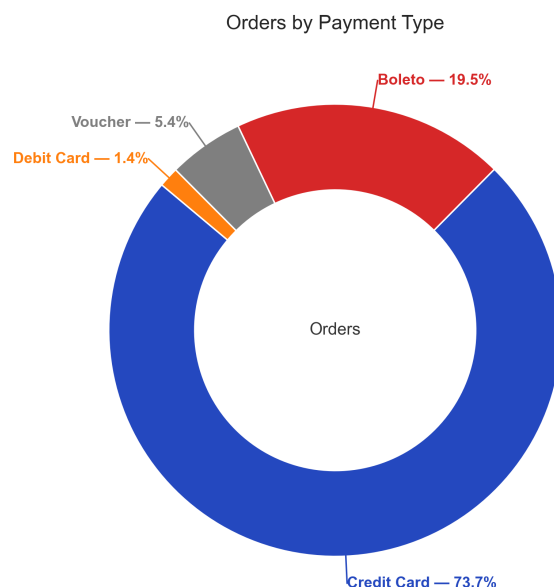


Figure 4: Distribution of total orders by payment type.

Note: The payment_type value “boleto” refers to a Brazilian bank slip (invoice) that can be paid at banks, via online banking, or at authorized payment locations. It functions similarly to a bill payment and is commonly used for offline payments.

The chart above shows the overall payment mix across the marketplace. Below is the payment-type share specifically within the top 10% highest-value orders.

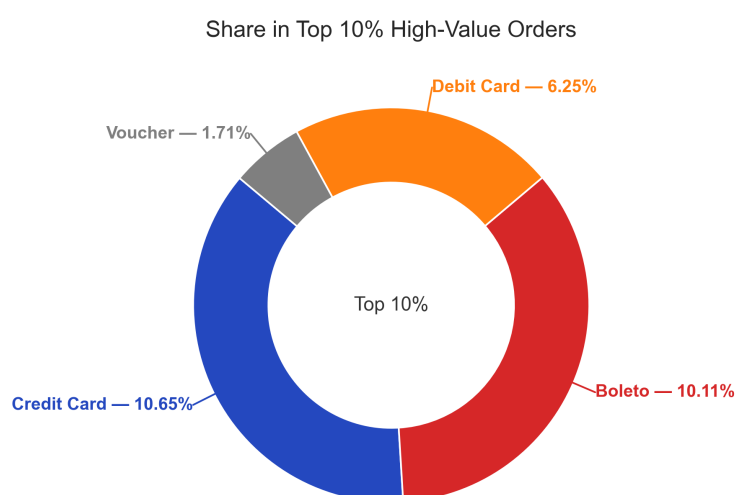


Figure 5: Payment-type share within the top 10% highest-value orders.

Insight: Credit card is the dominant payment method, representing approximately 73% of orders and nearly 77% of revenue. Boletto is the second most common method but at a much smaller scale. Voucher and debit card make up only a small share of transactions. The distribution of high-value orders across payment types is broadly proportional, indicating payment method is not a primary driver of order value.

Installment Behavior and Its Impact on Purchasing Patterns

To better understand how payment flexibility influences customer behavior, two complementary analyses were conducted: (i) the relationship between product price and the average number of installments by category, and (ii) the contribution of each installment plan to total order volume and revenue.

Product Price vs. Number of Installments by Category Figure 6 presents a bubble scatterplot comparing the average product price (x-axis), the average number of installments (y-axis), and the total number of orders per category (bubble size). Most categories form a concentrated cluster, indicating similar purchasing behavior across the majority of the catalog.

Two categories clearly stand out: *small appliances home oven and coffee* and *computers*. These segments combine both the highest product prices and the highest average number of installments. This pattern suggests that higherpriced products naturally lead consumers to rely more heavily on installment plans to complete the purchase.

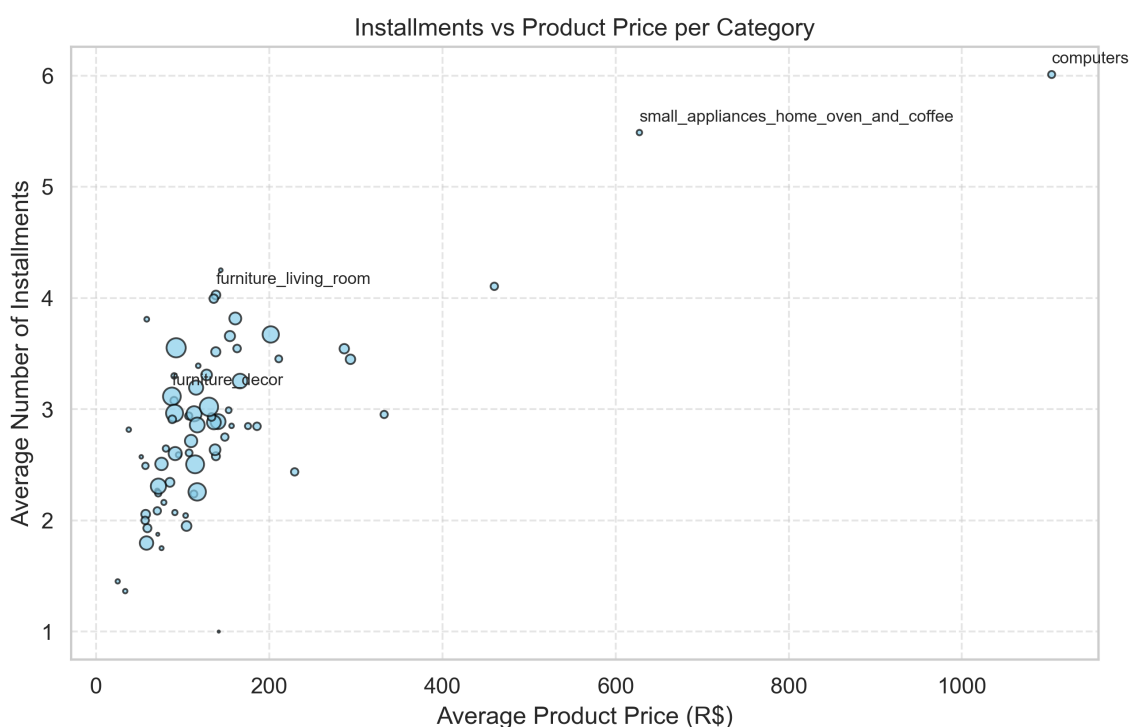


Figure 6: Average product price vs. average number of installments by category. Bubble size represents total order volume.

Order Volume vs. Revenue Share by Number of Installments To complement this view, Figure 7 compares the share of total orders with the share of total revenue generated by each installment plan. The

distribution reveals a clear asymmetry between purchasing frequency and financial contribution.

While single-installment payments represent nearly half of all orders (approximately 50%), they account for only 38% of total revenue, indicating these transactions are typically lower in value. In contrast, installment plans of 8 and 10 payments together represent only about 10% of all orders but contribute more than 22% of total revenue. This pattern underscores the strategic importance of installment options for enabling and capturing higher-value purchases.

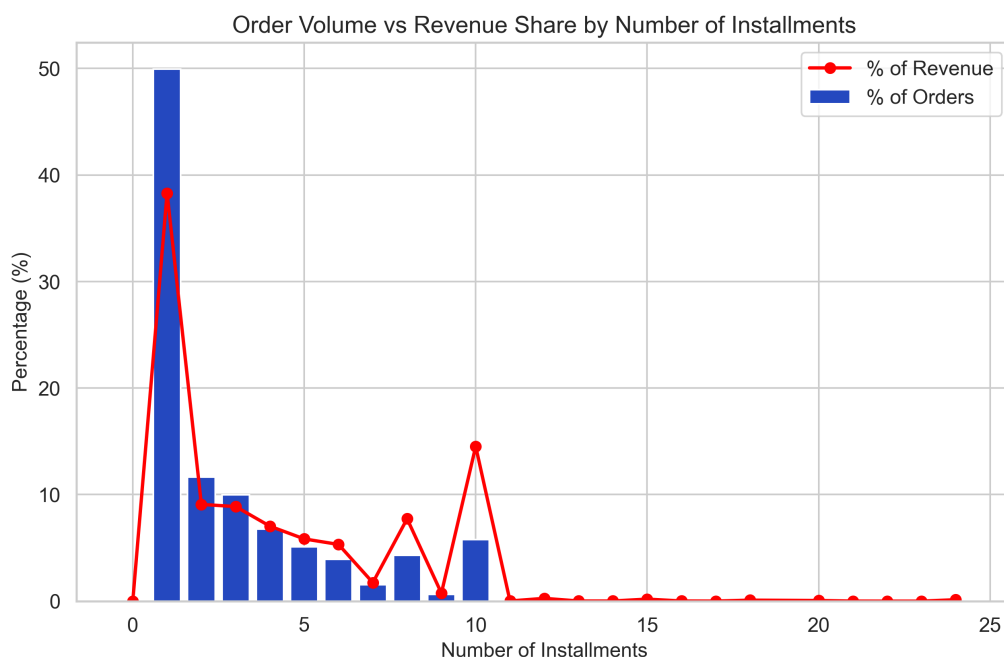


Figure 7: Comparison between order volume and revenue share by number of installments.

Insight Installments play a pivotal role in facilitating high-value purchases. While single-installment orders dominate in volume, multi-installment plans - especially 8 and 10 - disproportionately drive revenue. Categories with higher price points consistently rely on longer installment plans, reinforcing the importance of offering flexible payment options to support marketplace growth and conversion.

4.3 Customer & Behavior

The marketplace customer base is geographically concentrated: São Paulo (SP) accounts for approximately 42% of unique customers while representing roughly 22% of Brazil's population. This indicates a notable concentration of customers in São Paulo, likely reflecting higher purchasing power and marketplace penetration in the country's wealthiest state.

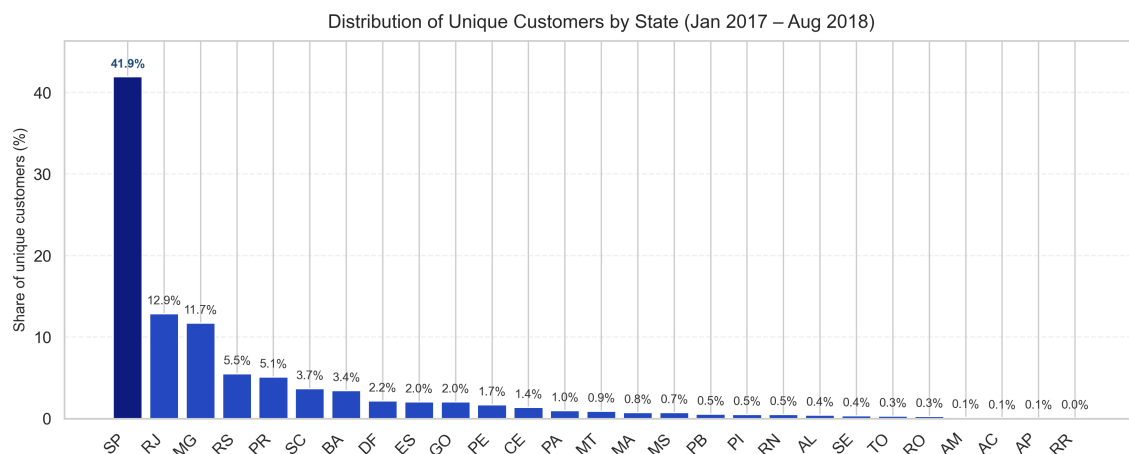


Figure 8: Share of unique customers by state (Jan 2017–Aug 2018). São Paulo (SP) highlighted.

Retention snapshot: Out of the **99k** orders in the analysis window, there are **95.8k** unique customers, implying only **3.1%** of customers placed more than one order. This extremely low repeat-purchase rate is the single most important commercial signal in the dataset: it points to a large opportunity to design retention and second-purchase initiatives.

Temporal purchase patterns: the heatmap below summarizes order volume by weekday and hour (7 × 24 grid). Darker cells indicate times of higher ordering activity. The marketplace shows clear peak hours and weekday patterns — useful inputs for customer communications, support staffing, and logistics scheduling.

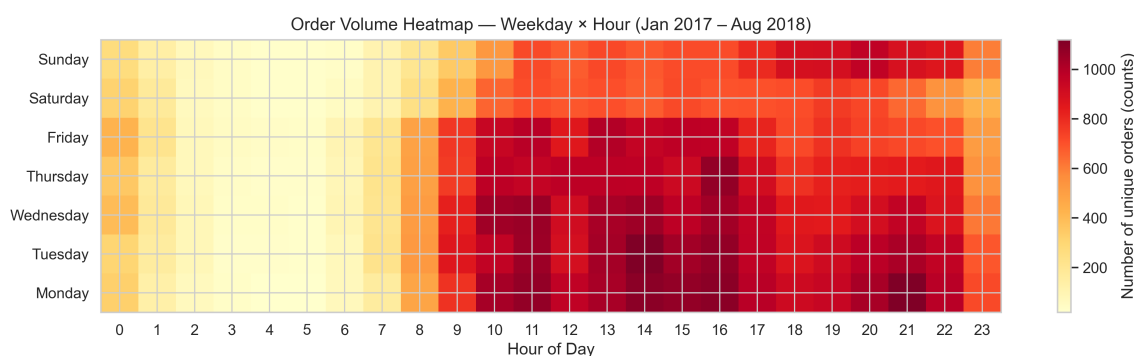


Figure 9: Orders heatmap (weekday × hour). Darker colors = more orders.

A closer look at the heatmap shows clear concentration patterns in customer activity. The darkest cells appear consistently on Monday and Tuesday, confirming that these are the busiest days of the week. Activity thins out as the week progresses, reaching its lowest levels on Saturday. Across hours of the day, three clusters stand out: late morning (around 11 AM), mid-afternoon (around 4 PM), and evening (around 9 PM). These peaks align with typical moments when customers are likely to take breaks or be available to browse online. Combined, the visualization reinforces the EDA findings: customers tend to shop primarily during weekdays and concentrate their purchases during daytime and early-evening hours, with afternoons showing the strongest demand.

Insight: São Paulo concentrates the largest share of customers, which aligns with its population size and purchasing power. The remaining customer base is distributed across other states in proportions that roughly mirror Brazil's population distribution. Overall, the marketplace shows a geographically balanced demand pattern, with no meaningful shifts in concentration over time.

Crucially, however, the analysis reveals an extremely low retention rate (3%), which stands out as the most significant finding in this section. Customer acquisition is occurring at scale, but it is not converting into repeat purchases — a major strategic gap.

A complementary look at the day-and-hour heatmap shows that customer demand follows predictable weekly and hourly cycles, with concentrated peaks rather than random fluctuations. This consistency suggests an opportunity: timing-based interventions, such as targeted campaigns during peak hours or incentives in low-traffic periods. These could help improve engagement, **but they do not address the core retention issue**, which is driven by factors beyond purchase timing.

4.4 Delivery Performance

This section evaluates the efficiency of Olist's delivery operations, focusing on timeliness, freight costs, and geographic disparities in shipping efficiency. The results highlight a strong logistics backbone, but also reveal important regional and commercial implications driven by freight expenses.

Delivery Timeliness

Overall delivery performance is highly positive. The majority of orders (**89.1%**) arrive earlier than the promised date, with customers receiving their products an average of **11.8 days in advance**. Only **6.59%** of orders are delivered late, indicating operational reliability across most shipments.

Across the full dataset, the average purchase-to-delivery time is **12.5 days**, encompassing both fulfillment and transportation intervals. These results suggest that shipping speed is a competitive strength for the platform.

Freight Cost Analysis

Freight plays a significant role in purchasing dynamics. The average freight cost per order during the analysis period is **R\$ 22.82**, accounting for **14.22%** of total revenue. This proportion is substantial and underlines the importance of shipping costs for both customer behavior and margins.

Freight costs also vary considerably across states, reflecting Brazil's large geographic dispersion and heterogeneous logistics infrastructure. These differences translate into meaningful variation in customer spending patterns: states with lower freight costs tend to show lower Average Order Values (AOVs), while states with higher freight costs display higher AOVs but lower overall sales volume.

To better understand these dynamics, Table 1 summarizes AOV, average product price, and average freight by state. The results confirm that both the product mix and freight intensity shape regional AOVs.

Customer State	AOV Total	Avg Product Price	Avg Freight
PB	263.39	191.71	42.75
AC	242.84	173.73	40.07
AP	239.16	164.32	34.01
AL	233.82	181.52	35.90
RO	227.50	165.97	41.07
RR	224.08	157.49	43.73
PA	222.83	165.61	35.85
TO	219.12	157.53	37.25
PI	218.84	160.27	39.15
RN	210.01	157.07	35.70
SE	209.47	153.50	36.78
CE	206.07	153.34	32.69
MT	205.38	148.41	28.08
MA	202.12	145.94	38.32
PE	194.86	145.31	32.87
MS	190.15	142.63	23.37
AM	188.08	135.50	33.21
BA	180.84	134.51	26.37
GO	172.33	126.28	22.77
SC	167.60	124.50	21.48
RJ	165.52	124.82	20.96
DF	164.96	125.75	21.05
RS	161.94	120.17	21.73
MG	159.65	120.82	20.64
ES	159.55	121.72	22.03
PR	158.89	119.28	20.56
SP	141.88	109.63	15.15

Table 1: Average Order Value, Product Price, and Freight Cost by State

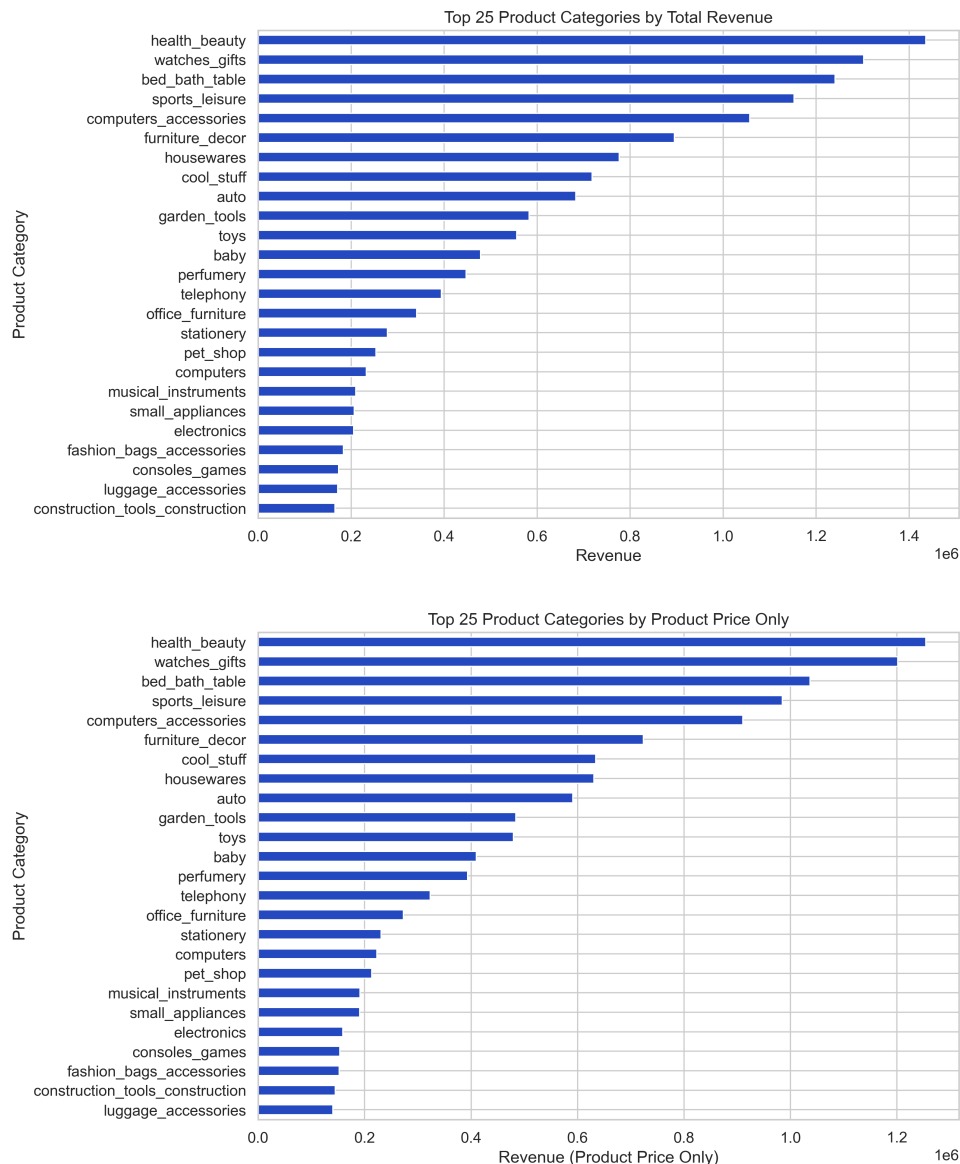
Category Analysis: Price vs. Total Revenue

Two complementary analyses were conducted to evaluate whether freight meaningfully influences category rankings. First, the top 25 product categories were ranked by total revenue including freight. Next, the same ranking was repeated using only the product price (excluding freight).

The results were nearly identical across both methods. This indicates that:

- freight costs are relatively uniform across product categories, and
- category revenue is driven primarily by product price rather than logistics cost.

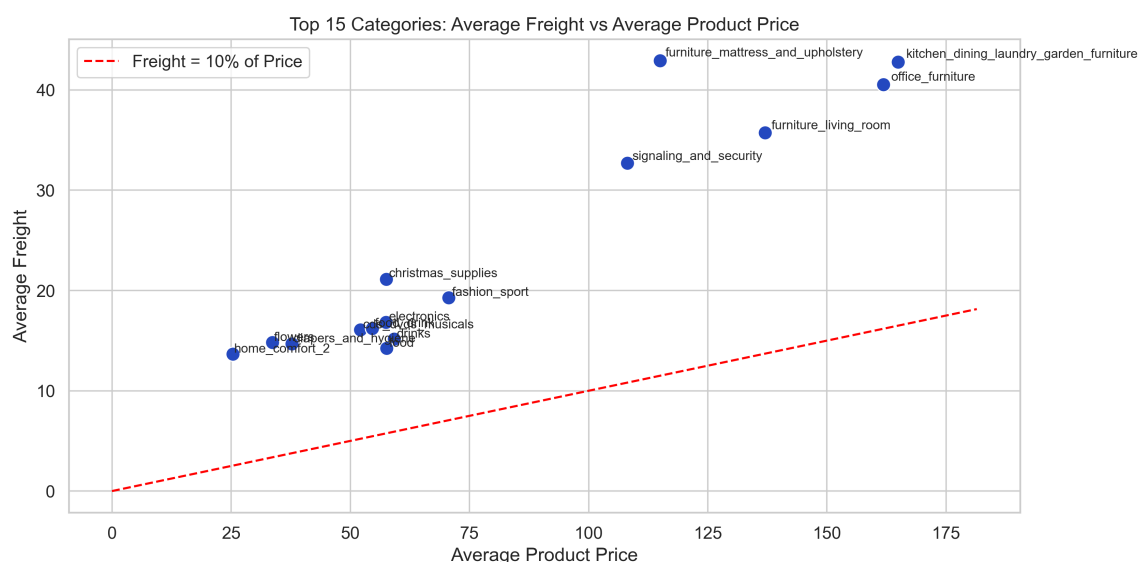
This consistency simplifies future category-based evaluations: analyses of category performance can safely rely on product price alone without risk of distorting the revenue structure.



Freight Costs Relative to Product Price

A complementary analysis was conducted to examine how freight cost compares to the value of products across categories. Two visualizations were created: (i) the freight-to-price ratio for all categories, and (ii) the top 15 categories with the highest freight-to-price ratios. These charts provide a clearer view of whether shipping disproportionately affects specific product groups.

The results show that freight cost is generally stable relative to item value for most categories, with only a few notable deviations. High-priced categories such as *[computers]* and *[small appliances home oven and coffee]* exhibit the lowest freight-to-price ratios, meaning that shipping represents only a small portion of the total purchase value. This reinforces earlier observations that these same categories also rely more heavily on installment payments—together indicating that they function as high-ticket segments where both financing and favorable logistics conditions play an important role in enabling purchases. Conversely, several furniture-related categories display significantly higher ratios, even without particularly low product prices, suggesting that in these segments shipping cost can disproportionately



Insight: Delivery performance is a clear operational strength of the marketplace: nearly 90% of orders arrive earlier than promised, and only 6.6% are late. However, freight emerges as a strategic lever with significant commercial impact. Shipping costs represent 14.2% of total revenue and vary substantially across states, shaping customer spending behavior. Regions with higher freight costs tend to show higher average order values but lower total revenue, suggesting that expensive delivery discourages frequent, lower-value purchases. Optimizing freight pricing in these areas could stimulate repeat buying and increase overall revenue.

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strategy and dashboard design.

A notable exception is the *computers* category, which, despite selling relatively few units, combines high product value with low shipping burden. This makes it a logistically efficient and strategically attractive category where targeted commercial efforts could yield outsized returns without additional operational strain.

4.5 Seller Analysis

Geographic Distribution of Sellers

The marketplace exhibits a pronounced geographic concentration of sellers, with the Southeast region dominating activity. São Paulo alone accounts for **53.55%** of all sellers, representing more than half of the entire marketplace. Minas Gerais (14.40%), Paraná (10.67%), and Rio de Janeiro (8.36%) follow at distant levels, while all remaining states individually represent less than 6% of the seller base.

This spatial imbalance mirrors the economic geography of Brazil and its existing logistics infrastructure. Sellers are concentrated in industrialized, densely populated regions, benefiting from shorter delivery routes and lower operational costs.

Market Concentration and Sales Inequality

Sales contribution is highly concentrated among top-performing sellers:

- Top 1% of sellers handle **26.19%** of all orders;
- Top 5% handle **52.22%**;
- Top 10% handle **66.44%**.

This means that two-thirds of all orders are processed by just one-tenth of the seller base. Such inequality is typical of large digital marketplaces and suggests the presence of established, professionalized sellers with optimized operations, extensive catalogs, and streamlined logistics networks.

Seller Location and Delivery Reliability

Delivery performance varies significantly across seller states. Regions closer to major logistics hubs consistently outperform northern or remote areas. For example, Amazonas (AM) and Maranhão (MA) exhibit some of the highest late-delivery rates, while Rio Grande do Sul (RS), Santa Catarina (SC), Goiás (GO), and Pernambuco (PE) achieve superior punctuality.

Southeastern states—particularly São Paulo—display strong reliability but still maintain a small share of “never arrived” orders (approximately 2%). Northern and isolated states depend on longer transportation routes and fewer distribution centers, resulting in substantially higher late-delivery percentages.

Overall, delivery reliability is strongly tied to seller geography and distance from dense logistics infrastructure.

Seller Review Scores

Customer review scores are relatively stable across different sales tiers:

- Overall average: **3.94**
- Top 1% sellers: **3.92**
- Top 5% sellers: **3.97**
- Top 10% sellers: **3.96**
- Top 20% sellers: **4.02**

Higher sales volume does not guarantee superior reviews. Review outcomes appear more strongly influenced by operational factors such as delivery punctuality and customer service than by seller size or business volume. Interestingly, the highest average ratings are found among the top 20%, not the highest-volume sellers, suggesting that mid- to high-volume sellers may balance scale and quality more effectively.

Catalog Breadth vs. Performance

Sellers were segmented into four groups based on catalog size:

Catalog Size	Avg Sales	Avg Review	Seller Count
Very Few SKUs	8.21	3.88	1846
Small Catalog	46.15	4.04	829
Medium Catalog	174.90	4.01	348
Large Catalog	871.29	3.95	45

The results show that while larger catalogs significantly increase sales volume, they do not offer a clear advantage in customer ratings. Specialized sellers with few SKUs achieve lower sales but maintain review scores that are not dramatically different from those of large-catalog sellers. Catalog expansion therefore appears to drive commercial output, but not customer satisfaction.

Key Insights and Implications

- **Geographic influence:** Seller location has a substantial impact on delivery reliability. Remote states experience higher late rates, while Southern and Southeastern sellers benefit from stronger logistics infrastructure.
- **High market concentration:** A small group of sellers dominates marketplace performance, with the top 10% processing nearly 70% of all orders.

- **Logistics as a driver of customer satisfaction:** Poor reviews correlate more strongly with delivery delays than with seller size or catalog length. Improving logistical reliability is likely the most effective way to enhance customer experience.
- **Catalog breadth drives sales, not ratings:** Larger catalogs increase sales volume but do not yield better ratings. Customer satisfaction is more closely tied to operational execution than to product variety.

Implications: Strengthening logistics - particularly for sellers in northern and inland regions - would likely improve customer satisfaction more effectively than expanding catalog size or increasing the number of sellers. Supporting smaller sellers through operational tools, fulfillment partnerships, or streamlined delivery solutions may help reduce concentration and improve overall marketplace performance.

4.6 Customer Reviews

Customer reviews offer a direct view into customer satisfaction and highlight how product quality, logistics performance, and shopping experience shape customer perception. The review dataset contains five possible scores (1 to 5), with the following distribution:

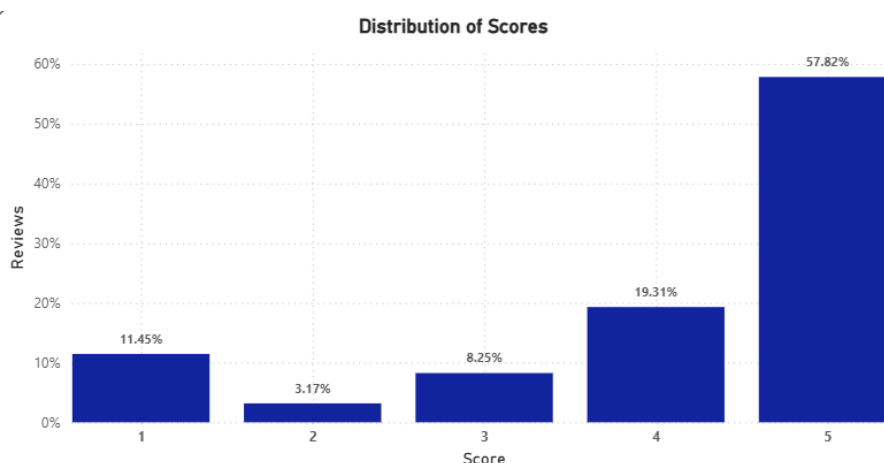


Figure 10: Distribution of Customer Review Scores

The majority of customers give positive feedback (scores 4 and 5), but a meaningful portion (about 11.5%) leave extremely negative evaluations, emphasizing that negative experiences are highly salient and strongly reflected in scores.

Review Scores by Product Category

Review scores are generally consistent across categories, with most averaging between 3.5 and 4.5. Two categories, however, stand out negatively:

- **Diapers and Hygiene** (avg. 3.21)
- **Security and Services** (avg. 2.50)

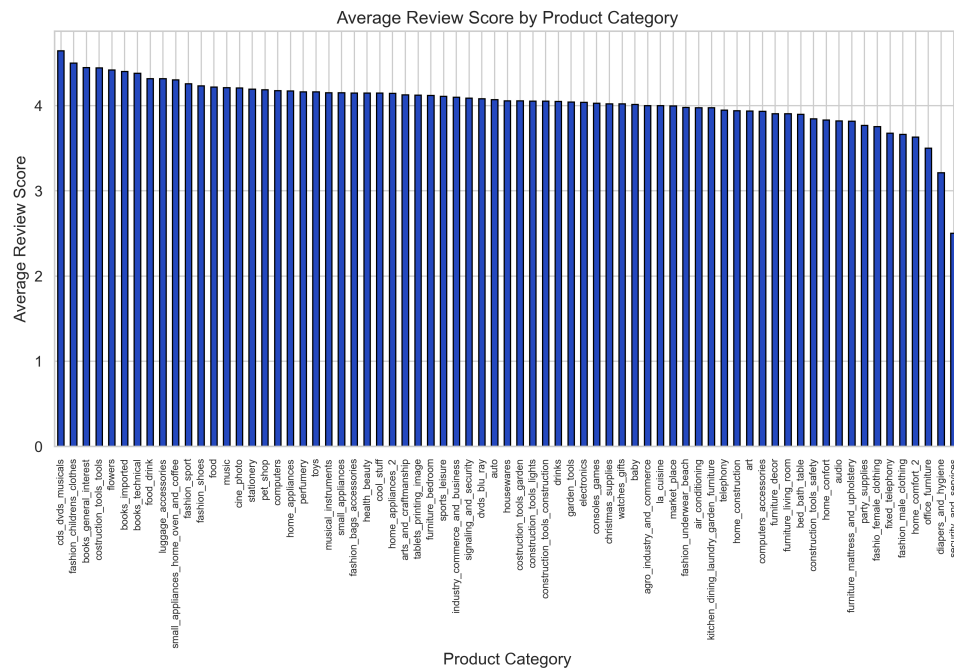


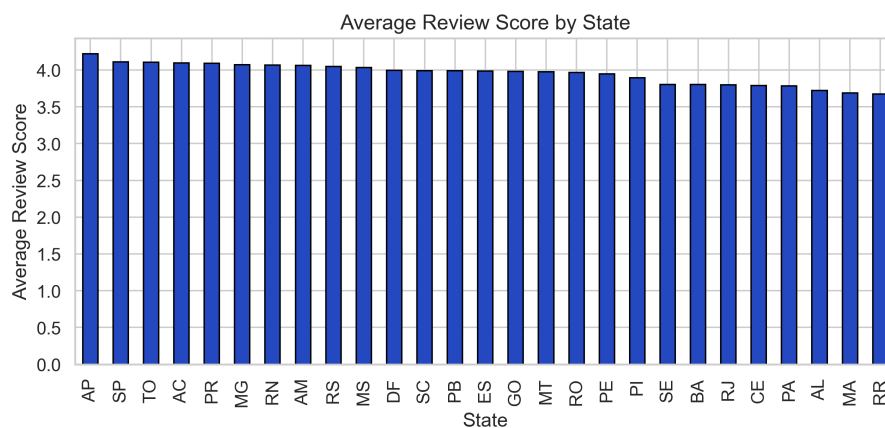
Figure 11: Average Review Score by Product Category

These significantly lower scores suggest issues related to product quality, mismatch of expectations, or service inconsistency. On the positive end, *cds_dvds_musicals* scores the highest (avg. 4.64).

Importantly, no clear relationship was observed between product price and review score: both low and high-priced products receive similarly distributed evaluations, indicating that price alone is not a primary driver of customer satisfaction.

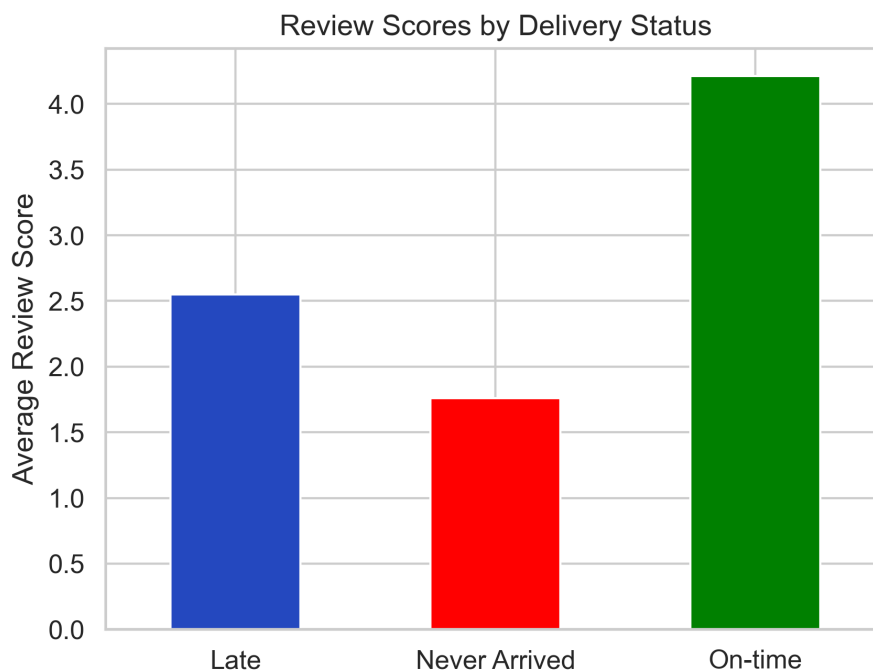
Review Scores by State

Average review scores show minimal geographic variation. Amapá (**AP**, 4.22) and São Paulo (**SP**, 4.11) score slightly above the national average, while Roraima (**RR**, 3.67) presents the lowest state average. Overall, state-level differences are small, suggesting that regional factors do not substantially influence review outcomes.



Delivery Performance and Review Scores

Delivery timeliness emerges as one of the strongest drivers of customer satisfaction. The relationship is direct and substantial:



- **On-time deliveries:** average score of **4.21**
- **Late deliveries:** average score drops sharply to **2.55**
- **Never arrived:** lowest score at **1.76**

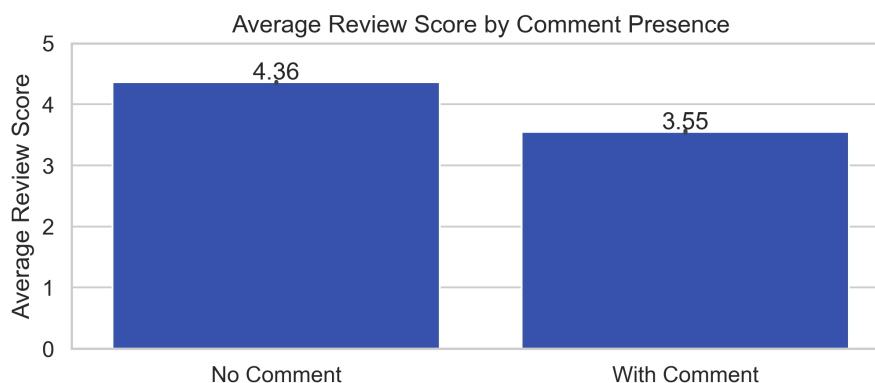
Given that approximately 10% of orders arrive late or do not arrive at all, even this relatively small proportion is enough to create a meaningful number of 1- and 2-star reviews. This reinforces the central strategic role of reliable logistics and delivery transparency in maintaining high marketplace reputation.

Written Reviews

About 42.1% of all reviews include a written comment. These comments are disproportionately associated with negative experiences:

- **Average score with written comment:** 3.55
- **Average score without comment:** 4.36

This indicates that dissatisfied customers are significantly more motivated to provide detailed feedback. Mining these reviews for themes such as delay complaints, product defects, packaging issues, or customer service problems offers a valuable opportunity to identify and address pain points.



Insight: Customer satisfaction is broadly stable across product categories, price ranges, and states, but two factors strongly differentiate the review experience: (1) specific underperforming categories (notably diapers, hygiene, and services), and (2) delivery performance. The latter stands out as the most critical driver of satisfaction: late or failed deliveries lead to dramatic declines in review scores and disproportionately generate written complaints. Strengthening logistics reliability and proactively addressing delay-related issues represents the clearest opportunity to improve overall customer satisfaction.

4.6.2 Review Text Analysis

To complement the quantitative review analysis, a text-based exploration was conducted using written customer comments. Reviews were divided into three score buckets: **Low** (1–2), **Neutral** (3), and **High** (4–5). Natural language processing techniques were applied to identify the most frequent and distinctive terms, as well as common bigrams and trigrams in each group.

Unigrams. Low reviews are dominated by words such as “*nao*” (not), “*recebi*” (received), “*produto*” (product), “*comprei*” (bought), and “*ainda*” (yet). These terms consistently signal problems related to non-receipt, delayed receipt, or issues in the delivery process. High reviews, in contrast, contain almost none of these complaint-related terms.

Bigrams. The most recurrent bigrams in low reviews include: “*nao recebi*” (did not receive), “*recebi produto*” (received product), “*ainda nao*” (not yet), and “*nao entregar*” (did not deliver). These patterns reinforce the centrality of failed or delayed deliveries as the primary driver of dissatisfaction.

Trigrams. The trigram analysis further highlights delivery failures as the key issue. Examples include: “*nao recebi produto*” (did not receive product), “*ainda nao recebi*” (did not receive yet), “*produto nao entregar*” (product not delivered), and “*produto nao chegar*” (product did not arrive). Customers often use repetitive phrasing to emphasize unresolved delivery problems.

Overall Insight Across unigrams, bigrams, trigrams, and comparative frequency analysis between low and high reviews, one theme dominates: **negative sentiment is overwhelmingly driven by delivery failures and delays**. Issues such as orders not arriving, uncertainty about delivery status, and lack of commu-

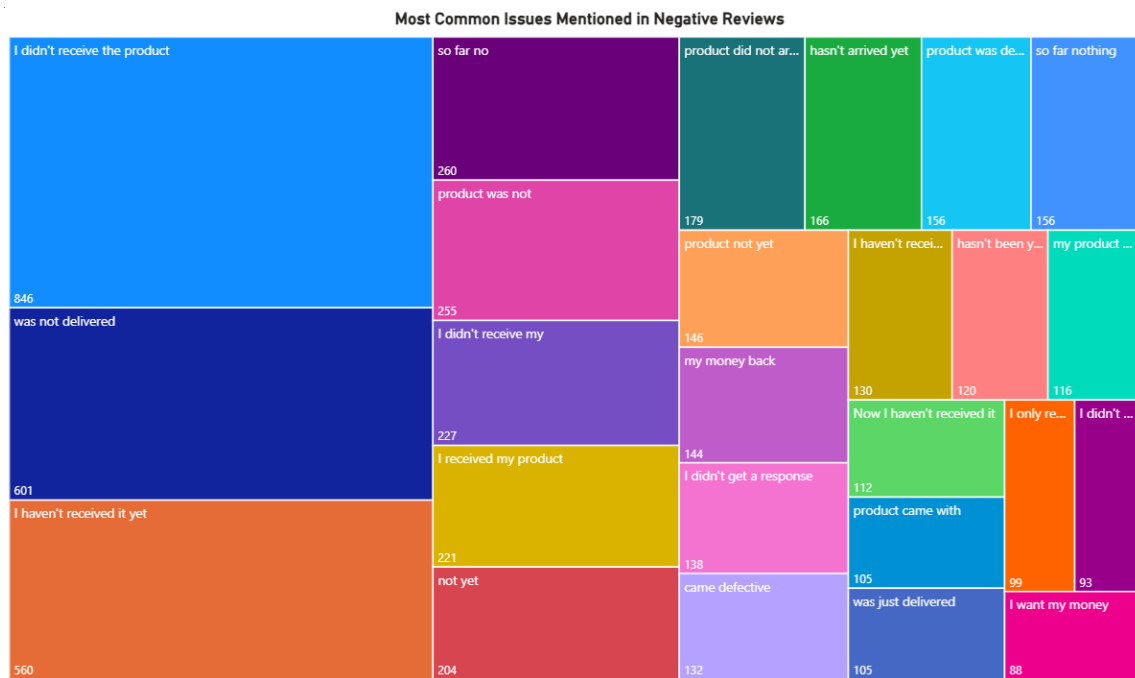


Figure 12: Most frequent negative trigrams in low-scored reviews (top 25). Size and number indicate occurrence count.

lication clearly contribute to the majority of low ratings (≤ 2). Improving logistical reliability—especially shipment tracking, punctuality, and proactive communication—represents the most direct opportunity to reduce negative reviews and elevate overall customer satisfaction.

4.7 Product Presentation: Descriptions and Photos

To assess whether product presentation influences customer behavior, three complementary analyses were conducted: (i) the correlation between product description length, number of photos, and units sold; (ii) the distribution of photo counts across products; and (iii) sales patterns among products with unusually high numbers of photos.

Correlation Between Presentation Attributes and Units Sold

A correlation matrix was computed using the variables *product_description_length*, *product_photos_qty*, and *units_sold*. The resulting coefficients, displayed in Table 2, show no meaningful linear relationship between product presentation attributes and sales volume. Description length exhibits essentially zero correlation with units sold (-0.01), while number of photos shows only a weak negative relationship (-0.06).

To visualize these relationships, Figure 13 presents scatterplots of units sold against description length and photo count, respectively. Both exhibit broad dispersion with no clear trend, reinforcing the absence of an underlying relationship.

	Description Length	Photo Count	Units Sold
Description Length	1.00	0.12	-0.01
Photo Count	0.12	1.00	-0.06
Units Sold	-0.01	-0.06	1.00

Table 2: Correlation matrix for product presentation attributes and units sold.

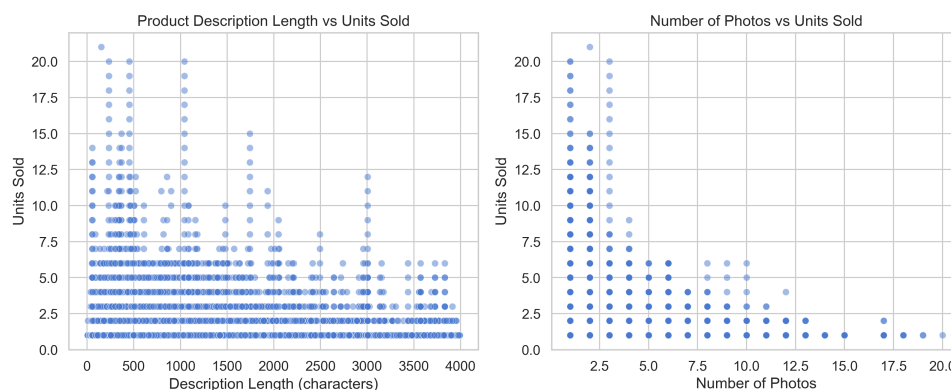


Figure 13: Product description length vs. units sold.

Distribution of Product Photo Counts

Table 3 summarizes the distribution of the number of photos per product. The majority of products include only 1–3 photos, with progressively fewer products offering more extensive visual documentation. Products with more than 10 photos represent a very small fraction of the catalog.

Sales Patterns Among Products with High Photo Counts

To investigate whether extensive visual documentation influences sales, an additional analysis focused on the top 10% of products by photo count. Figure 14 shows units sold versus number of photos among products with more than five photos.

The scatter remains widely dispersed, with no observable upward trend. Even among products with unusually high photo counts, additional imagery does not appear to increase sales.

Insight **Product presentation does not significantly influence sales performance.** Neither description length nor the number of photos shows a meaningful correlation with units sold. Most products include only 1–3 photos, and even products with unusually many photos do not sell more. This suggests that, in this marketplace, product presentation features alone are not key drivers of purchase behavior, and other factors - such as price, delivery, and seller performance - likely play a more substantial role in determining sales outcomes.

Photo Count	Number of Products
1	55,870
2	21,872
3	12,346
4	8,406
5	5,356
6	3,771
7	1,498
8	718
9	313
10	341
11	68
12	53
13	30
14	6
15	12
17	11
18	4
19	2
20	1

Table 3: Distribution of the number of photos per product.

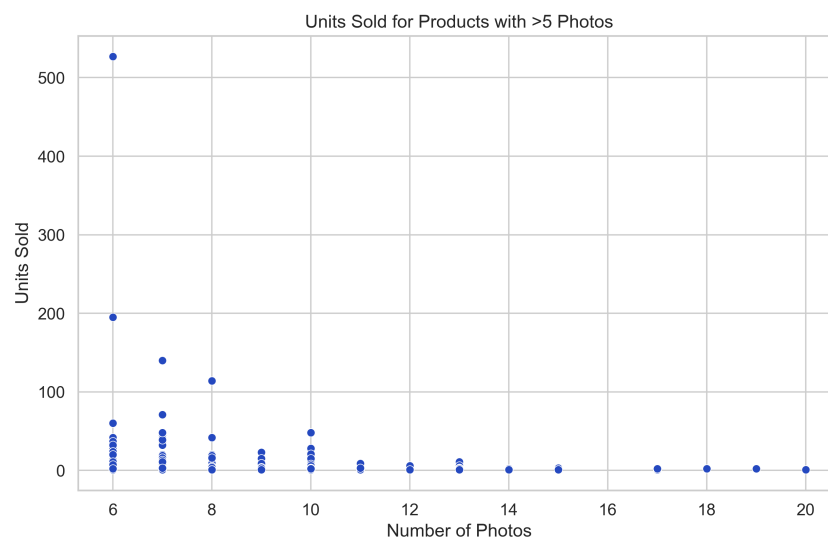


Figure 14: Units sold vs. number of photos for products with more than five photos.

5 Power BI Dashboard Overview

This project includes an interactive Power BI dashboard designed to provide a comprehensive view of customer behavior, seller performance, delivery operations, product dynamics, and payment patterns within the Olist marketplace dataset.

The dashboard can be accessed at the following link: [Click Here](#)

All pages of the dashboard share a consistent navigation structure and filtering capabilities. Each page includes:

- A **date range filter** that allows the user to restrict the view to any period;
- A **state filter**, enabling geographic analysis across Brazil;
- A **product category filter**, for category-level exploration;
- A **payment method filter**, supporting comparison of payment behaviors over time.

These filters apply dynamically across all visuals within each page, enabling flexible and targeted exploration of marketplace dynamics.

Dashboard Structure

The dashboard is composed of five pages, each focused on a different analytical pillar.

1. Overview

This page provides a high-level summary of the marketplace, including total orders, revenue, customer distribution, and delivery timeliness. It functions as a central hub highlighting broad performance patterns.

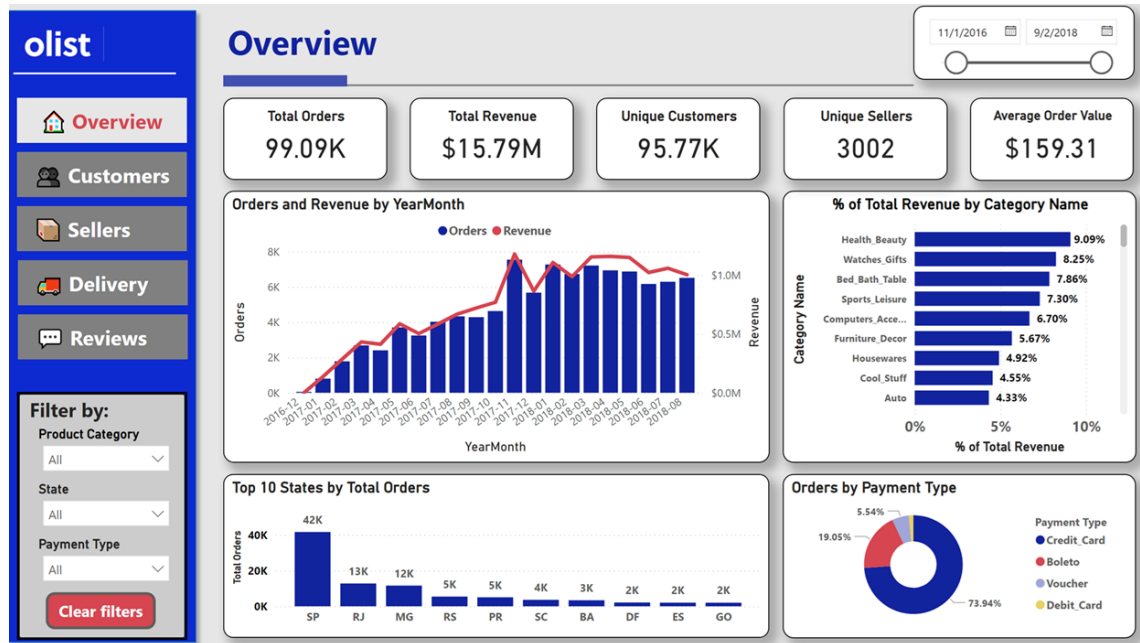


Figure 15: Dashboard Overview Page

2. Customers & Behavior

This page examines customer demographics, order patterns, retention rates, purchasing cycles, and differences across states and categories. It includes heatmaps, trend charts, and retention indicators.

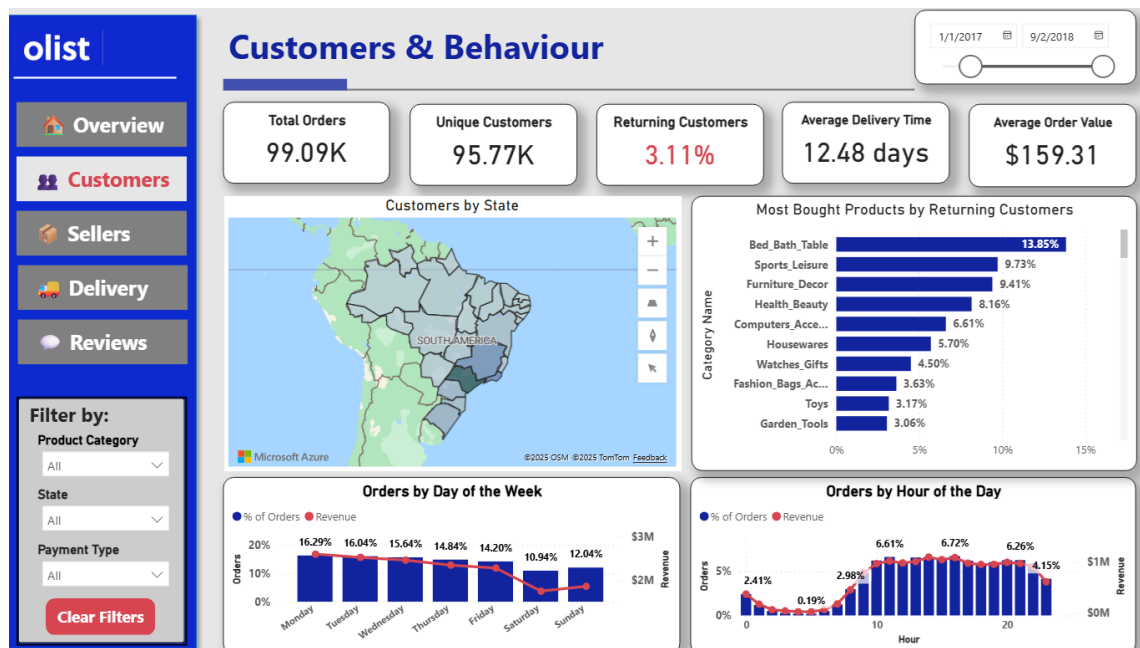


Figure 16: Customers and Behavior Page

3. Sellers

The Sellers page focuses on geographic distribution, market concentration, seller catalog size, and the relationship between seller logistics and delivery outcomes. It also highlights the contribution of top-

performing sellers.

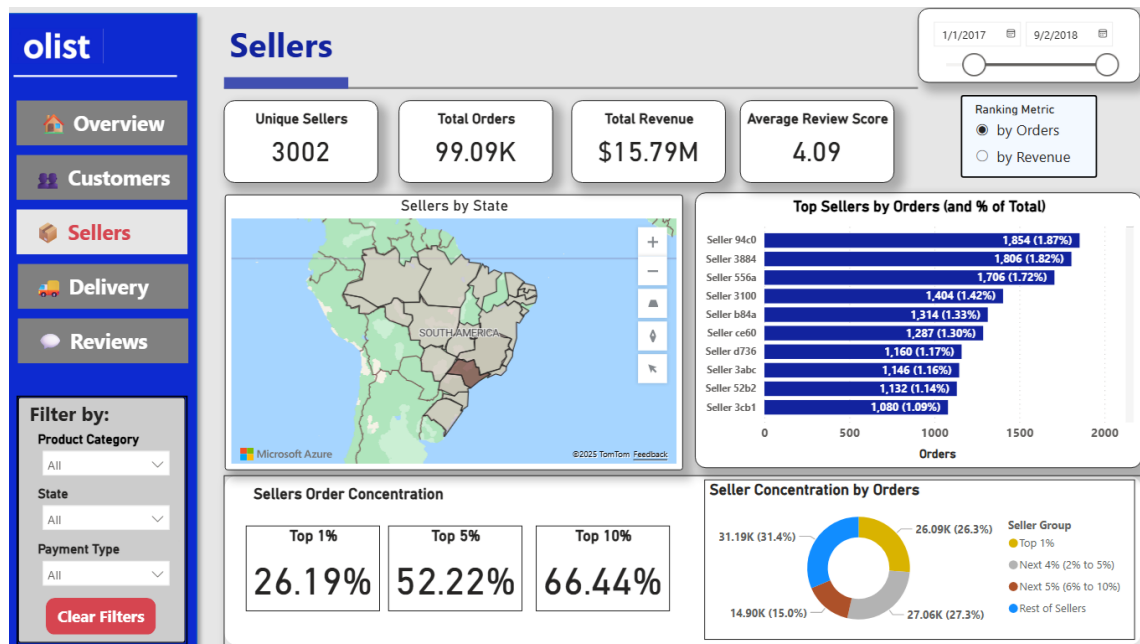


Figure 17: Sellers Page

4. Delivery Performance

Focusing on logistics, this page visualizes on-time delivery rates, delays, geographic delivery variation, and relationships between freight cost, revenue, and delivery speed.

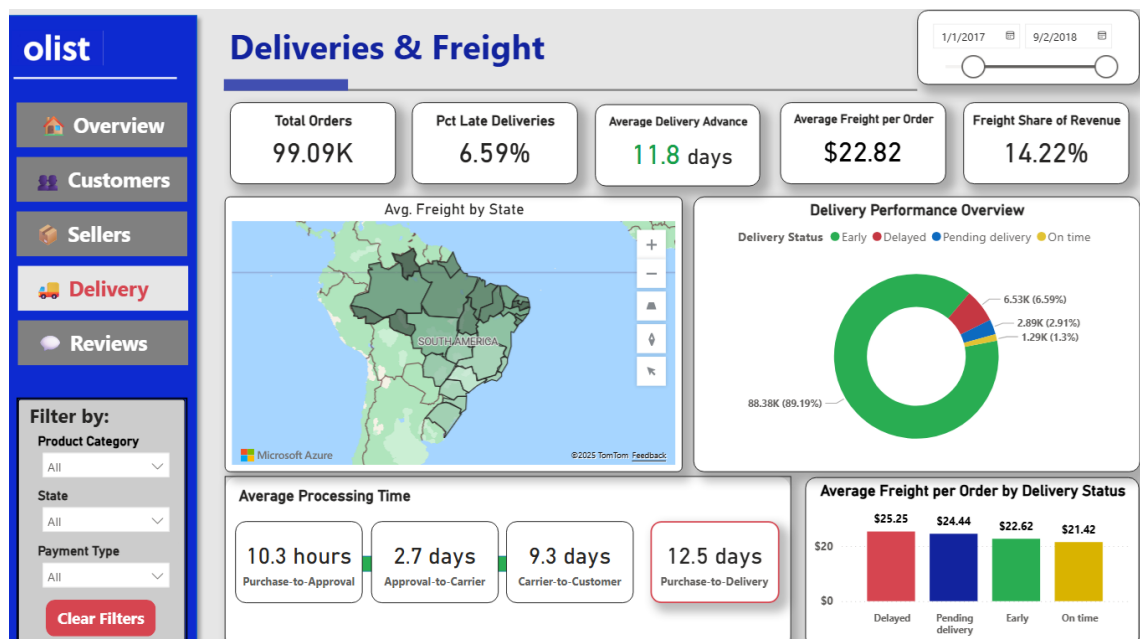


Figure 18: Delivery Performance Page

5. Reviews

The Reviews page summarizes customer satisfaction through score distributions, category comparisons, sentiment indicators, and connections between delivery timeliness and review outcomes.

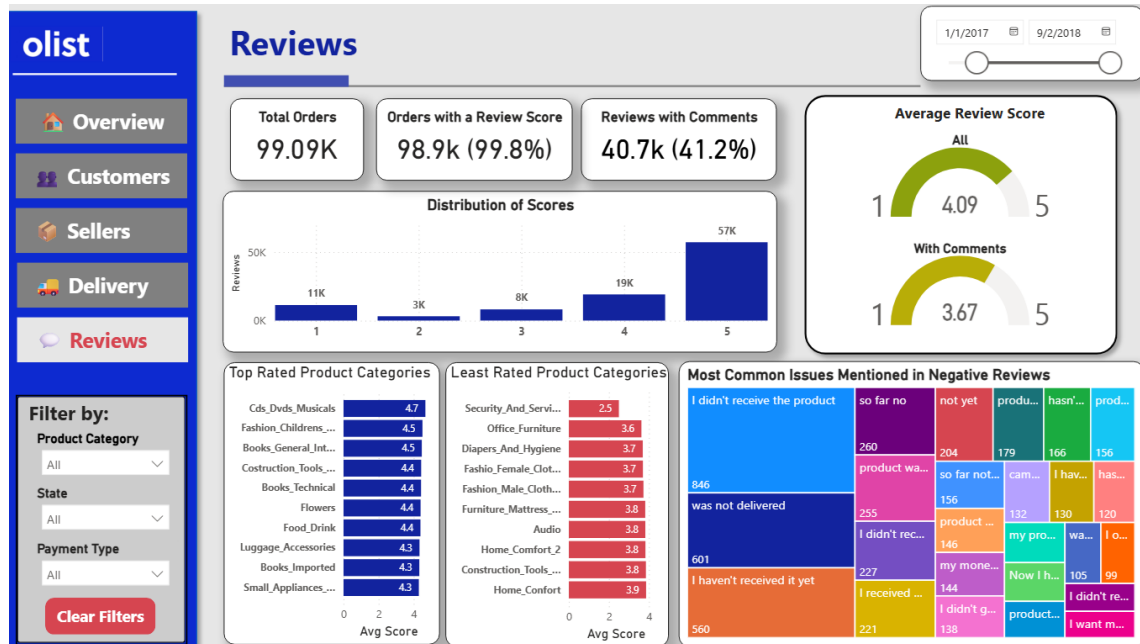


Figure 19: Reviews Page

Overall, this dashboard consolidates the insights explored throughout Section 4 and translates them into an interactive analytical environment. Its design supports both strategic decision-making and operational monitoring across the marketplace.

6 Final Summary and Recommendations

This report presents a comprehensive exploratory analysis of the Olist e-commerce dataset, covering customer experience, delivery performance, seller behavior, product presentation, geographic patterns, and temporal dynamics. Together, these analyses reveal clear patterns about what drives satisfaction, retention, and operational efficiency in the marketplace.

6.1 Overall Insights

1. Customer Retention Is Critically Low

Retention analysis reveals the single most important commercial insight in the dataset: out of approximately 99,000 orders, there are 95,800 unique customers, meaning only about 3.1% of customers placed more than one order during the analysis window. This extremely low repeat-purchase rate indicates that Olist operates primarily as a one-time transaction platform rather than a relationship-based marketplace. This represents a major opportunity for developing targeted retention, reactivation, and second-purchase strategies that could substantially increase lifetime value and reduce acquisition dependence.

2. Delivery Performance Drives Customer Satisfaction

Delivery punctuality is the strongest predictor of review scores:

- On-time deliveries (approximately 89.7%) correspond to an average rating of 4.21.
- Late deliveries (approximately 7.6%) correspond to an average rating of 2.55.
- Never-arrived orders (approximately 2.7%) correspond to an average rating of 1.76.

Late and failed deliveries are directly tied to negative reviews, churn risk, and repeated complaints such as “nao recebi”, “ainda nao recebi”, and “produto errado”.

3. Geographic Disparities Affect Fulfilment

Customers are concentrated in SP, RJ, and MG, while sellers are overwhelmingly concentrated in SP. States such as AM and MA show significantly higher late delivery rates, indicating regional logistics challenges and the need for differentiated service-level strategies.

4. Seller Dynamics: Concentrated but Not Performance-Driven

Marketplace activity is highly concentrated:

- The top 1% of sellers handle approximately 26% of all orders.
- The top 10% of sellers handle approximately 68% of all orders.

However, top sellers do not deliver better customer satisfaction; their average ratings remain around 3.9–4.0. Reliability, rather than scale, is the primary driver of positive outcomes: low-rated sellers tend to have higher rates of late or failed deliveries.

5. Product Presentation Does Not Influence Sales Volume

Correlation and scatterplot analyses show that:

- Product description length does not meaningfully correlate with units sold.
- Number of product photos does not meaningfully correlate with units sold.

Most products have only one to three photos, and even those with more than five photos do not achieve higher sales. Product presentation is not a key sales driver in the Olist marketplace context.

6. Freight-to-Price Dynamics Influence Customer Behavior

Some categories, particularly furniture and bulky items, have disproportionately high freight-to-price ratios, which can deter purchases and reduce conversion rates. This effect is especially visible for sellers located outside SP.

7. Text Analysis Confirms Logistics Pain Points

TF-IDF, bigram, and trigram analyses show that negative reviews overwhelmingly focus on logistics issues, missing items, incorrect quantities, invoice problems, and poor support responsiveness.

8. Temporal Trends Show Event-Driven Demand Surges

There is a massive spike on 2017-11-24 (Black Friday), representing the highest order volume in the entire dataset. Event-aware capacity planning is crucial for logistics and seller operations.

6.2 Final Recommendations

1. Develop a Customer Retention and Second-Purchase Strategy

The extremely low repeat-purchase rate (only about 3.1% of customers place more than one order) represents a major commercial opportunity. To address this, Olist should:

- Implement targeted post-purchase campaigns to encourage second purchases (e.g., personalized discounts, product recommendations, reminders).
- Build a CRM-based lifecycle strategy with segmentation for new, dormant, and high-value customers.
- Introduce a loyalty or rewards program focused on increasing customer lifetime value.
- Monitor repurchase intervals and design automated triggers for win-back flows.

2. Strengthen Logistics and Reduce Late Deliveries

- Expand or optimize regional hubs in the North and Northeast regions.
- Improve SLA agreements with carriers in states showing high late rates.
- Provide sellers with packaging and dispatch guidelines to standardize operations.

3. Improve Tracking and Proactive Customer Communication

- Implement predictive alerts for potential delays.
- Provide clearer and more frequent tracking updates to reduce support volume.

4. Implement Seller Performance Management

- Monitor KPIs such as on-time rate, never-arrived rate, return rate, and review distribution.
- Offer operational playbooks and mandatory training for low-performing sellers.
- Reward high-reliability sellers through visibility boosts or category prioritization.

5. Optimize Marketplace Experience During Retail Events

- Temporarily scale logistics capacity.
- Allocate dedicated customer support squads.
- Run seller readiness and inventory planning campaigns.

6. Deploy Voice-of-Customer Monitoring

- Automate alerts for frequent complaint patterns such as “nao recebi” and “produto errado”.
- Prioritize operational fixes based on complaint volume and customer impact.

7. Rethink Product Presentation Guidelines

Since product description length and number of photos do not influence sales, marketplace guidelines can focus on minimum acceptable standards rather than quantity or length. This allows operational factors to take priority over cosmetic improvements.

Closing Note

This analysis provides a strong foundation for improving customer experience, operational efficiency, and seller performance across the Olist marketplace. The insights and recommendations developed here can inform future dashboards, monitoring tools, and predictive models, helping transform raw data into strategic action.